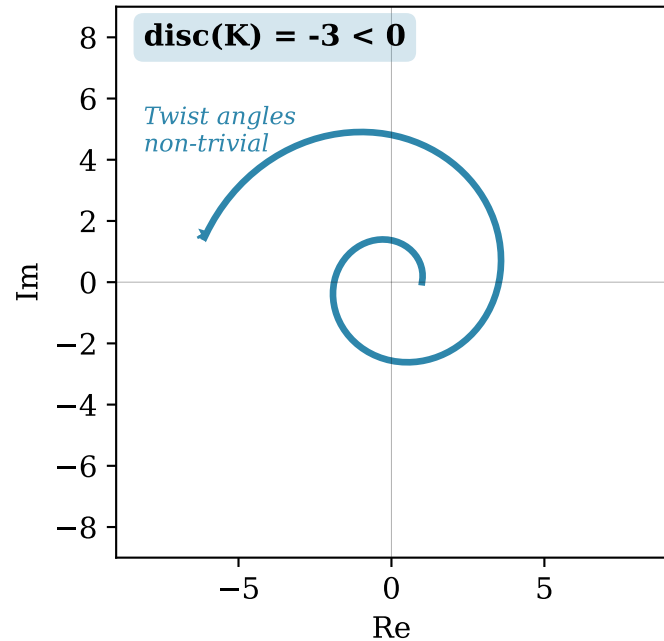


Trace field dichotomy: geometric origin of lepton/quark CP asymmetry

$K_{\text{PMNS}} = Q(\sqrt{-3})$ [imaginary]

Twisted loxodromic holonomy

$\delta_{\text{CP}} = 195.91^\circ$ (large)



$K_{\text{CKM}} = Q(\sqrt{17})$ [real]

Purely hyperbolic holonomy

$J_{\text{CKM}} = 0$ (zeroth order)

